

SUMOTO, Japan

WMO: 477760

Lat: 34.3105N

Lon: 134.8490E

Elev: 69

StdP: 100.50

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	0.9	-7.4	2.0	1.8	-6.1	2.3	2.6	8.7	5.8	7.5	4.7	3.6	300	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	32.0	25.6	31.0	25.3	29.9	25.0	26.5	30.3	26.0	29.5	25.5	28.7	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.8	28.6	24.9	20.2	28.1	24.5	19.6	27.6	83.2	30.6	80.8	29.6	78.8	28.8	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.4	5.4	-1.7	33.5	1.0	1.0	-2.5	34.2	-3.1	34.8	-3.7	35.4	-4.4	36.1		
			-3.9	27.3	1.1	0.7	-4.6	27.9	-5.3	28.3	-5.9	28.7	-6.7	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	5.3	5.7	8.8	13.9	18.6	22.0	26.1	27.4	23.9	18.6	13.1	8.0	(6)
(7)		DBStd	8.00	2.16	2.58	3.03	3.04	2.28	2.14	2.08	1.69	2.48	2.54	2.90	2.69	(7)
(8)		HDD10.0	416	147	124	61	5	0	0	0	0	0	6	73	(8)	
(9)		HDD18.3	1723	404	355	296	136	26	1	0	0	27	157	321	(9)	
(10)		CDD10.0	2605	1	3	23	121	265	359	499	541	416	267	99	11	(10)
(11)		CDD18.3	870	0	0	0	2	32	111	241	282	166	36	1	0	(11)
(12)		CDH23.3	6126	0	0	0	4	65	368	1947	2751	936	54	0	0	(12)
(13)		CDH26.7	1541	0	0	0	0	4	42	497	834	162	2	0	0	(13)
(14)	Wind	WSAvg	2.5	3.0	2.8	2.8	2.6	2.4	2.1	2.3	2.2	2.3	2.4	2.5	2.9	(14)
(15)	Precipitation	PrecAvg	1397	47	65	104	107	152	181	153	109	180	145	100	56	(15)
(16)		PrecMax	2080	112	164	180	206	416	376	306	358	377	421	283	148	(16)
(17)		PrecMin	772	12	15	56	33	60	53	32	12	42	39	16	4	(17)
(18)		PrecStd	322	28	38	32	48	80	73	83	96	93	95	66	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	16.4	19.1	23.7	26.9	30.0	33.2	33.7	31.6	26.5	21.0	17.0	(19)
(20)			MCWB	11.0	14.3	13.8	16.3	18.7	24.2	26.3	25.6	25.3	22.7	18.3	14.5	(20)
(21)		2%	DB	10.9	13.6	17.0	21.6	25.0	27.7	31.6	32.4	29.8	24.8	19.6	14.2	(21)
(22)			MCWB	8.1	10.9	13.3	15.4	18.0	22.8	25.6	25.6	24.6	21.3	16.9	11.8	(22)
(23)		5%	DB	9.7	11.3	15.5	20.0	23.7	26.5	30.5	31.5	28.4	23.5	18.5	12.9	(23)
(24)			MCWB	7.0	8.5	12.2	15.3	17.7	21.9	25.4	25.4	23.9	20.2	15.9	10.2	(24)
(25)		10%	DB	8.6	9.7	14.0	18.7	22.5	25.4	29.4	30.5	27.3	22.3	17.5	11.9	(25)
(26)			MCWB	6.0	6.9	10.6	14.6	17.4	21.5	24.9	25.2	23.4	18.9	14.9	9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	14.8	16.2	19.2	21.7	25.1	27.3	27.2	26.2	23.8	19.2	15.2	(27)
(28)			MCD B	12.9	16.4	17.3	21.1	23.7	28.3	31.7	31.3	29.7	25.6	20.1	16.4	(28)
(29)		2%	WB	9.0	11.3	14.2	17.5	20.5	24.2	26.4	26.6	25.4	22.2	17.7	12.4	(29)
(30)			MCD B	10.1	13.1	16.2	19.7	22.5	26.5	30.3	30.5	28.3	24.2	19.0	13.8	(30)
(31)		5%	WB	7.5	9.1	12.6	16.3	19.5	23.4	25.8	26.1	24.8	20.9	16.4	10.9	(31)
(32)			MCD B	9.1	10.6	15.0	18.8	21.9	25.4	29.3	29.8	27.3	22.9	18.1	12.4	(32)
(33)		10%	WB	6.4	7.4	11.1	15.1	18.5	22.5	25.2	25.7	24.2	19.6	15.2	9.7	(33)
(34)			MCD B	8.3	9.3	13.7	18.0	21.3	24.6	28.3	29.1	26.5	21.7	17.1	11.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	5.7	6.8	7.4	7.1	5.8	5.6	6.0	5.3	5.0	5.0	5.0	(35)
(36)			MCD BR	6.3	7.5	8.6	8.9	8.6	7.2	6.9	7.1	6.4	5.7	5.6	5.9	(36)
(37)			MCWBR	5.1	6.0	6.5	5.4	4.7	3.7	3.3	3.1	3.3	3.9	4.8	4.9	(37)
(38)		5% WB	MCD BR	5.8	7.0	8.3	7.6	6.8	5.6	6.2	6.3	5.6	5.0	5.3	5.5	(38)
(39)			MCWBR	5.4	6.4	7.1	5.6	4.9	3.5	3.3	3.0	3.2	4.0	5.1	5.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.393	0.437	0.496	0.529	0.533	0.552	0.554	0.525	0.486	0.446	0.415	0.388	(40)	
(41)		taud	2.196	2.086	1.939	1.912	1.972	2.002	2.047	2.119	2.199	2.225	2.218	2.225	(41)	
(42)		Ebn at Noon	789	792	776	772	773	757	752	767	779	776	759	768	(42)	
(43)		Edh at Noon	114	142	178	192	183	177	168	154	136	123	112	106	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.47	3.08	4.05	4.85	5.23	4.68	5.10	5.26	4.09	3.33	2.57	2.22	(44)	
(45)		RadStd	0.17	0.23	0.29	0.45	0.52	0.46	0.64	0.48	0.36	0.33	0.22	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page